The relativity of wrong, redux

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Is it irrational to believe that God exists?

Is it irrational to hope for eternal life?

Has science made religion obsolete?

The positivist story (Auguste Comte)

Science	Newton, Darwin, Hawking
$\Uparrow$	
Philosophy	Aristotle, Descartes, Kant
$\Uparrow$	
Religion	Judaism, Christianity, Islam
$\Uparrow$	
Myth	Zeus, Loki

Science's victorious march

	Religion	Science	
2013	Soul	Neuroscience	
1860	Creation	Evolution	Darwin
1700	Miracles	Natural Laws	Newton
1600	Geocentric	Heliocentric	Galileo
1200	Creation	Eternal Universe	Aristotle
⋮	⋮	⋮	

What's wrong with these two stories about science

POINT: Most past scientific theories have been wrong

COUNTERPOINT: Some theories are wronger than others; science develops through "successive refinements" (Isaac Asimov: "The relativity of wrong")

What's wrong with these two stories about science

POINT: Science is not obviously progressive, not obviously converging to the truth

- Wine is good for you (2000). No, wine is bad for you (2008). No, wine is in fact good for you (2013).
- Personality is due primarily to genetics. No, it's due primarily to environmental conditions. No, genetics is the most important factor.

What's wrong with these two stories about science

- Science surrenders the territory it apparently gained

eternal universe	$\longrightarrow$	big bang
heliocentric	$\longrightarrow$	no preferred reference frame
natural laws	$\longrightarrow$	quantum mechanics

Case Study: The Eternity of the World

- Aristotle: Second great philosopher; first great scientist
- Rejects Plato's forms. The world consists of en-formed matter.
- Rejects Plato's claim that there can be no knowledge of the physical world $\longrightarrow$ made physics possible

Conflict areas between Aristotle and Religion

1. Materialism and the possibility of life after death.
2. Ethics
3. The creation of the universe

Aristotle's Physics

- In every change, something stays the same (underlying matter). What changes is the *form*.
- Every change is a sort of movement.
- Nature abhors a vacuum.

The Universe is Eternal

1. Suppose for *reductio ad absurdum* that the universe came into existence at some finite time. Then matter existed after that time, but didn't exist before that time. But the change (from non-existence to existence) requires an underlying substratum, and so matter existed before the universe existed. This is a contradiction.
2. Suppose for *reductio ad absurdum* that there was a first motion. Then either the universe existed eternally in a state of rest before the first motion, or the universe came into existence and then something moved. The second option is a contradiction, because the universe coming into existence would itself be an object moving. The first option is also wrong.

St. Bonaventure

- Reason was flawed by the fall, and cannot find truth without the help of faith. (Distrust of Aristotle)
- Reason, when used properly, shows that the universe is finite.
- Contemporary parallels:
  - Creation Science
  - Intelligent Design
  - Alvin Plantinga: Augustinian Science

Thomas Aquinas

- Criticism of Bonaventure: bad science reflects poorly on Christianity (compare Francis Collins on intelligent design)
- Science cannot demonstrate that the world is eternal, nor that it was created.
- "Was the world created?" Can only be answered by faith
- Reason cannot contradict faith, but Reason has fixed limits.
- Contemporary parallels
  - Still holds sway in many Catholic circles
  - Non-overlapping magesteria

Siger of Brabant and Boetius of Dacia

- Accused of holding the doctrine of **double truth**: two contradictory things can both be true

  As a scientist, I believe that the world is eternal.
  As a Christian, I believe that the world is not eternal.

- A scientist can only admit those things that follow the principles of his science
- Science cannot admit creation events
- As far as science is concerned, every natural event has a natural cause
- Fideism: some things should be believed even though not demonstrable (???)
- Contemporary parallels
  - Methodological naturalism

If those guys were alive today . . .

- Bonaventure: reject mainstream science, develop a Theistic alternative
- Aquinas: show where science leaves off and faith begins
- Siger and Boetius: science is a serious game

The missing point of view

None of our medieval thinkers entertains the idea that science could answer *all* questions (contrast with Dawkins, Dennett, Harris, etc.)

For Aquinas, no strict separation between Reason (Science) and Faith. Aquinas calls Theology a Science. But distinguish:

- Things that Reason can prove (e.g. God exists).
- Things that are matters of faith (e.g. Jesus is the incarnate God, the universe was created in time).

Scientism: Science can answer all questions

- What is science? Is sociology a science? Is history a science? Is philosophy a science?

  > *"For Quine, philosophical theories are to be assessed in the ways that scientific hypotheses are, and good philosophical method is continuous with good scientific method."*

- Does science tell me what I ate for breakfast this morning?
- Does science tell me that it's wrong to kill?
- Does science tell me that a Brahms concerto is beautiful?

The Galileo Affair

- Christian Doctrine before 1600: the earth is the center of the universe (geocentrism)
- Geocentrism fits the data only at the price of horrendous complications (epicycles)
- The Copernican revolution: everything becomes simple if we put the sun in the center
- Galileo condemned for teaching heliocentrism

They were all wrong

- Does the earth go around the sun, or the sun around the earth?
- Galilean relativity: No such thing as absolute rest
- General relativity: No such thing as absolute rotation

  There is no objective right answer about whether A is rotating around B or vice versa.
- Lesson: The debate (X or not-X) was based on a false presupposition.

Prospective views on science and religion

- Secular scientism: Science and religion are in competition, and science wins
- Religious scientism: Science will eventually come around to the old truths of religion
- Separatism: Science and religion are non-overlapping magesteria (Steven Jay Gould). Science has to do with facts, religion with values.
- A science is a "rule governed activity" with limited objectives and limited possibilities.

  According to the principles of this science...